

# Fuel Tank Assembly Defect Rate Reduction

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# Business Case

## Background

- The Company assembles fuel tanks used in heavy-duty truck manufacturing. Tanks must meet requirements for leak integrity, surface condition, component configuration, and dimensional conformity before being released to the customer. Defective tanks require rework, additional inspection, or scrap and may interrupt production flow.

## Business problem

- During the baseline study, 9 of 50 inspected fuel tanks failed final inspection, resulting in an 18% defect rate. This is more than twice the project target of less than 8%.

## Business impact

- Additional labor for rework and inspection
- Material loss when tanks cannot be repaired
- Reduced first-pass yield and production throughput
- Risk of delayed delivery and customer quality concerns



# Business Case - Estimated Cost Benefit

## Current situation

- Monthly production: **6160 fuel tanks**
- Current defect rate: **18%**
- Target defect rate: **8%**
- Estimated cost per defective tank: **\$150**

## Expected improvement

- Current defects:  $6,160 \times 18\% = 1,108.8 \approx 1,109$
- Target defects:  $6,160 \times 8\% = 492.8 \approx 493$
- Defects avoided:  $1,109 - 493 = 616$  tanks/month

## Estimated savings

- $616 \times \$150 = \$92,400$  estimated monthly savings
- $\$92,400 \times 12 = \$1,108,800$  estimated annual savings
- *Illustrative estimate only. Monthly production volume and average defect cost must be verified using the Company production and cost records.*



# Problem & Goal Statement

**Y Metric:** Fuel Tank Assembly Defect Rate (%)

**Problem :** The current Fuel Tank Assembly Defect Rate is **18%**, resulting in rework, scrap, and reduced production efficiency.

**Goal:** Reduce the Fuel Tank Assembly Defect Rate from **18% to less than 8% by December 2026.**

# CTQs

| Customer requirement                   | CTQ                    | Specification/acceptance criterion                        | Measurement                             |
|----------------------------------------|------------------------|-----------------------------------------------------------|-----------------------------------------|
| Tank must retain fuel safely           | Leak integrity         | No detectable leakage during the approved pressure test   | Pass/fail pressure test                 |
| Tank must have acceptable appearance   | Surface condition      | No scratch or dent exceeding the approved visual standard | Visual inspection using defect examples |
| Tank must fit the truck assembly       | Dimensional conformity | All dimensions within drawing tolerance                   | Gauge or measurement record             |
| Components must be assembled correctly | Configuration accuracy | Correct brackets, fittings, caps, and orientation         | Configuration checklist                 |
| Tank must pass inspection initially    | First-pass quality     | Pass without rework                                       | Final-inspection record                 |

# Operational Definitions

| Item             | Operational definition                                                                                                |
|------------------|-----------------------------------------------------------------------------------------------------------------------|
| Unit             | One completed fuel tank presented for final inspection                                                                |
| Defective tank   | A tank with one or more confirmed defects at final inspection                                                         |
| Scratch          | Surface damage exceeding the approved visual acceptance standard                                                      |
| Leak             | Failure of the approved air-pressure or leak test                                                                     |
| Dent             | Permanent deformation classified as minor or major using the inspection standard                                      |
| Misconfiguration | Missing, incorrect, or improperly positioned component                                                                |
| Minor defect     | Repairable without scrapping or structurally replacing the tank                                                       |
| Major defect     | Defect requiring major repair, replacement, or scrap                                                                  |
| Recording rule   | Record date, shift, operator experience, fixture, material lot, result, defect type, severity, and rework requirement |
| Multiple defects | Count the tank once in the defective-tank rate but record every defect type separately                                |
| Formula          | $\text{Defective tanks} \div \text{total tanks inspected} \times 100$                                                 |

# DPMO and Sigma Level

## MEASURE

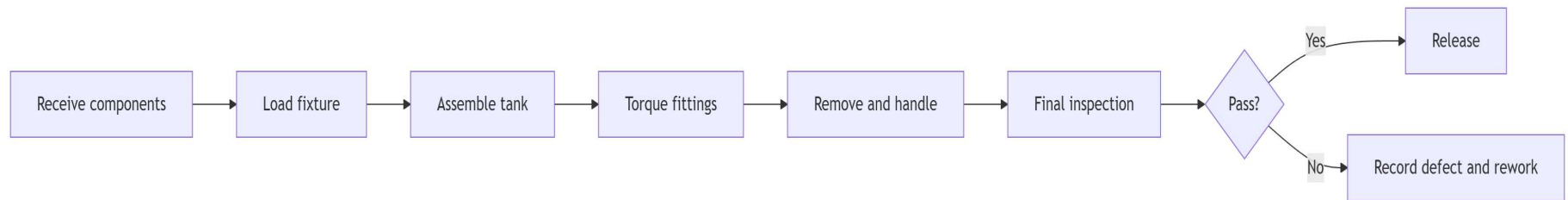
Using **one final quality opportunity per tank**:

- Units inspected = 50
- Defective units = 9
- Opportunities per unit = 1
- Total opportunities =  $50 \times 1 = 50$ 
  - $DPMO = \frac{9}{50 \times 1} \times 1,000,000 = 180,000$

Using the conventional 1.5-sigma shift:

- Sigma level =  $\text{NORMSINV}(1 - 0.18) + 1.5 \approx 2.42$

The baseline process operated at approximately 180,000 DPMO and a 2.42 sigma level. This indicates substantial opportunity to improve first-pass fuel-tank quality.



## Current State - Process Map

- Likely defect entry points:
  - Scratches: loading, fixture contact, removal, and handling
  - Leaks: seal installation, fitting assembly, and torque application
  - Dents: material handling and fixture contact
  - Misconfiguration: component selection and assembly verification

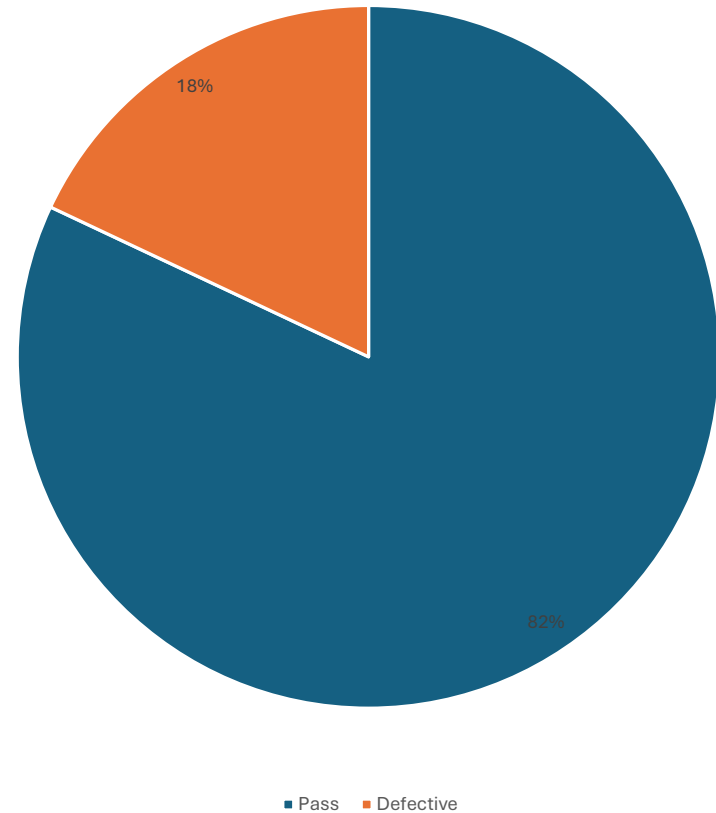


# Current State - Performance

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- 50 inspected
- 41 passed
- 9 defective
- First-pass yield = 82%
- Defect rate = 18%
- DPMO = 180,000
- Sigma level  $\approx 2.42$
- Scratch and leak = 66.7% of recorded defects

Baseline Fuel Tank Inspection Results



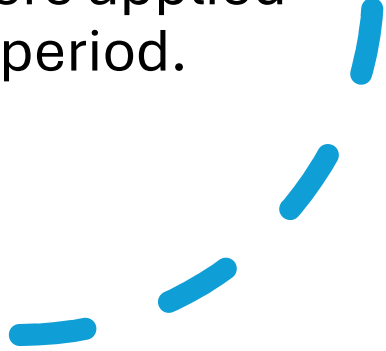


# Data Collection Plan

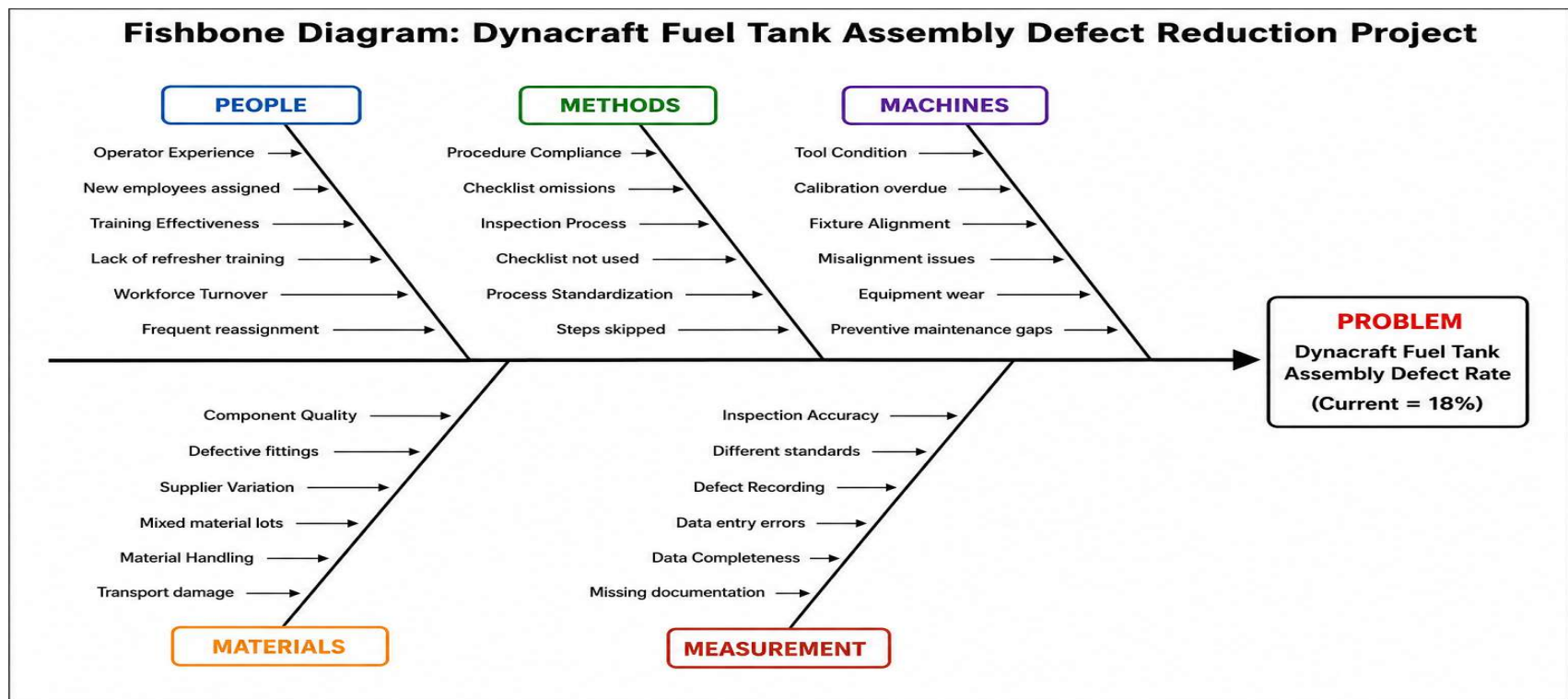
| Item              | Description                                       |
|-------------------|---------------------------------------------------|
| Y Variable        | Fuel Tank Defect Rate                             |
| Data Type         | Discrete                                          |
| Sample Size       | 50 Fuel Tanks                                     |
| Collection Period | June 22 – July 6                                  |
| Data Source       | Final Inspection Quality Log                      |
| Collected By      | Quality Inspector                                 |
| Related Variables | Shift, Operator Experience, Material Lot, Fixture |

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# Measure System

- All completed fuel tanks were inspected after assembly.
  - Each tank was classified as **Pass**, **Minor Defect**, or **Major Defect**.
  - Inspection results were recorded immediately in the production quality log.
  - The same inspection criteria were applied throughout the data collection period.
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- Four blue curved lines of varying lengths and orientations, located in the bottom right corner of the slide.

# Fishbone Diagram



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## Fishbone Diagram Analysis

The fishbone generated potential causes. The collected data showed the largest observed difference for operator experience, while fixture and shift were not statistically significant. These results identify areas for further investigation but do not confirm causation.

Four blue curved lines of varying lengths and orientations, arranged in a cluster in the bottom right corner of the slide.

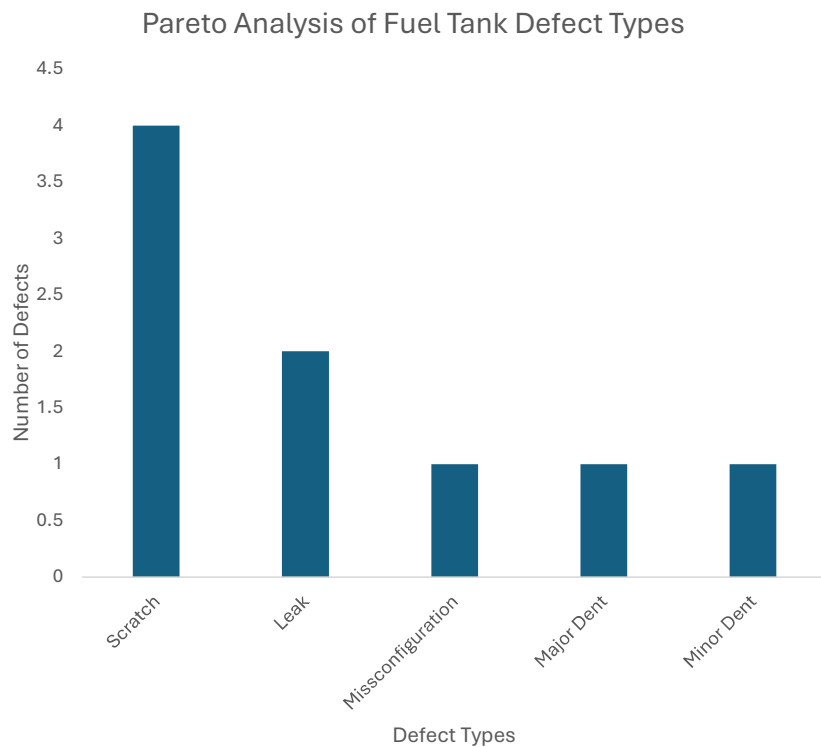
# Analysis – Defect Type Frequency

## Observed results

- **Scratch:** 4 of 9 defects (**44.4%**)
- **Leak:** 2 of 9 defects (**22.2%**)
- **Scratch and Leak combined:** 6 of 9 defects (**66.7%**)
- Misconfiguration, Major Dent, and Minor Dent each accounted for **11.1%**

## Takeaway

- Scratch and Leak were the most frequently recorded defect categories, accounting for 66.7% of the nine observed defects. They should receive priority in the X1 operator-training pilot and continued root-cause investigation.



# Analysis of X1 – Defect Rate by Operator Experience

X1: Operator experience

Y: Fuel Tank Assembly Defect Rate

## Observed results

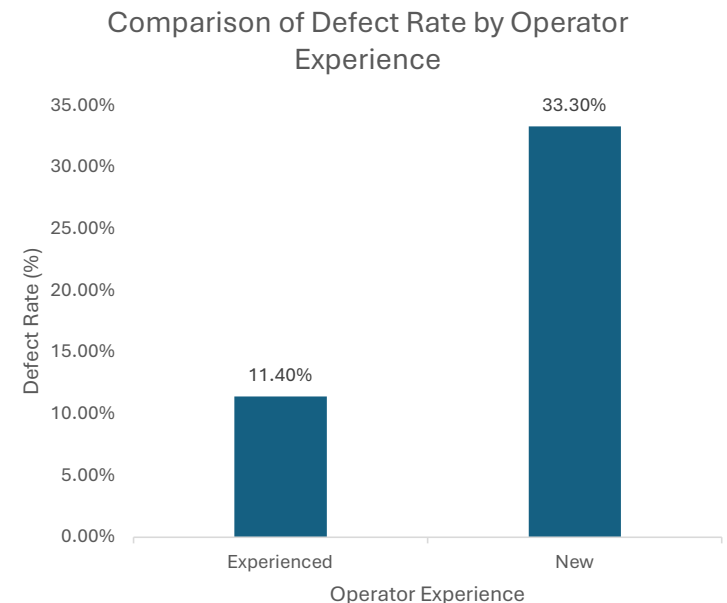
- New: 5/15 defective, or 33.3%
- Experienced: 4/35 defective, or 11.4%
- Risk Ratio =  $\frac{33.3\%}{11.4\%} = 2.92$

## Statistical test

- Fisher's exact test:  $p = 0.106$
- Because  $p > 0.05$ , the result is not statistically significant.

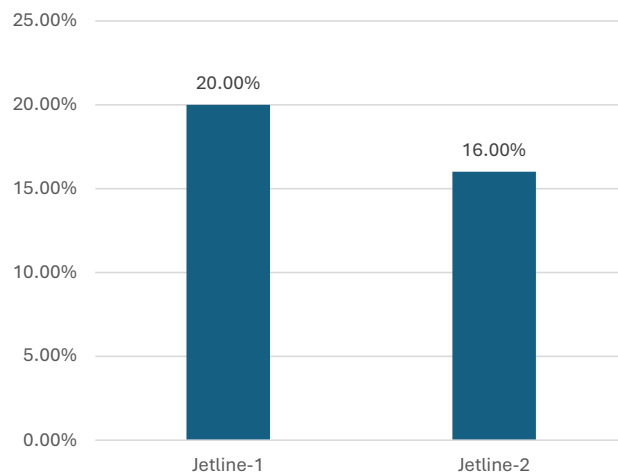
## Takeaway

- Operator experience showed the largest observed practical difference. However, a larger sample is required before confirming X1 as a root cause.



# Analysis – Defect Rate by Fixture

Comparison of Defect Rate by Fixture



X2: Fixture used

Y: Fuel Tank Assembly Defect Rate

Jetline-1: 20%

Jetline-2: 16%

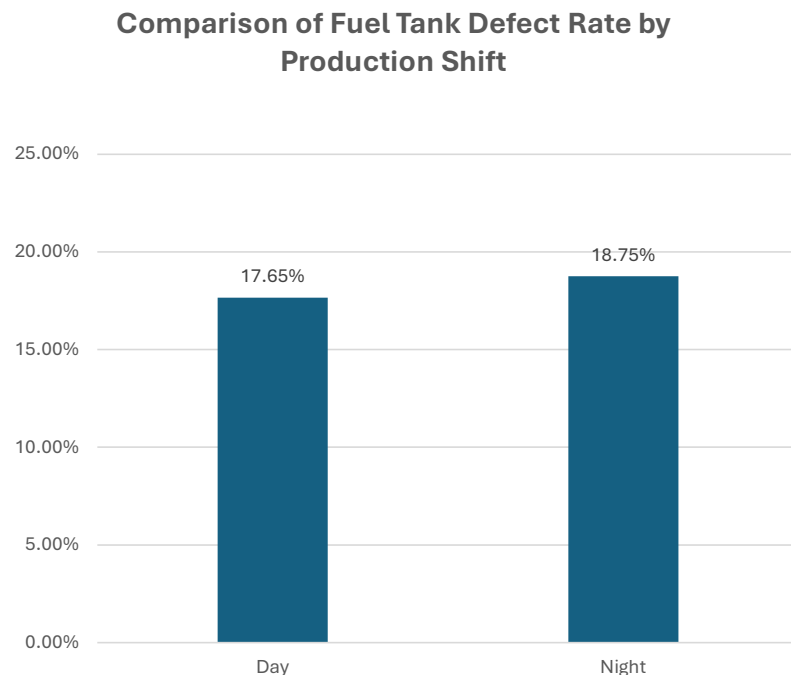
Fisher's exact test:  $p = 1.000$

## Conclusion

- Neither fixture nor shift showed a statistically significant association with defect occurrence.



# Analysis – Defect Rate by Shift



X3: Production shift

Y: Fuel Tank Assembly Defect Rate

Day: 17.65% Night: 18.75%

Fisher's exact test:  $p = 1.000$

## Conclusion

- X3 did not show a statistically significant relationship with the Y metric and is not supported as a primary root cause.

# Statistical Validation of Potential Factors

| Factor analyzed     | Defect-rate comparison               | Fisher's p-value | Interpretation                                                       |
|---------------------|--------------------------------------|------------------|----------------------------------------------------------------------|
| Operator experience | New: 33.3% vs<br>experienced: 11.4%  | <b>0.106</b>     | Largest observed<br>difference, but not<br>statistically significant |
| Fixture             | Jetline-1: 20% vs Jetline-<br>2: 16% | <b>1.000</b>     | No statistical evidence<br>of an association                         |
| Production shift    | Day: 17.65% vs night:<br>18.75%      | <b>1.000</b>     | No statistical evidence<br>of an association                         |

- At the 0.05 significance level, none of the evaluated factors showed a statistically significant association with defect occurrence. Operator experience had the largest practical difference and should be investigated with a larger sample.

## Analysis conclusion:

- Prioritize operator training as a pilot improvement based on the observed risk difference, while continuing data collection before confirming it as a statistically validated root cause.

# Analysis Limitations and Next Steps

## Limitations

- The analysis included only **50 tanks and 9 defective units**.
- Small group sizes produced limited statistical power.
- Results show associations, not confirmed causation.
- Material-lot effects were not evaluated.

## Next steps

- Collect 100–200 additional observations
- Continue tracking operator, fixture, shift, and material lot
- Record defect type and location
- Repeat the statistical analysis

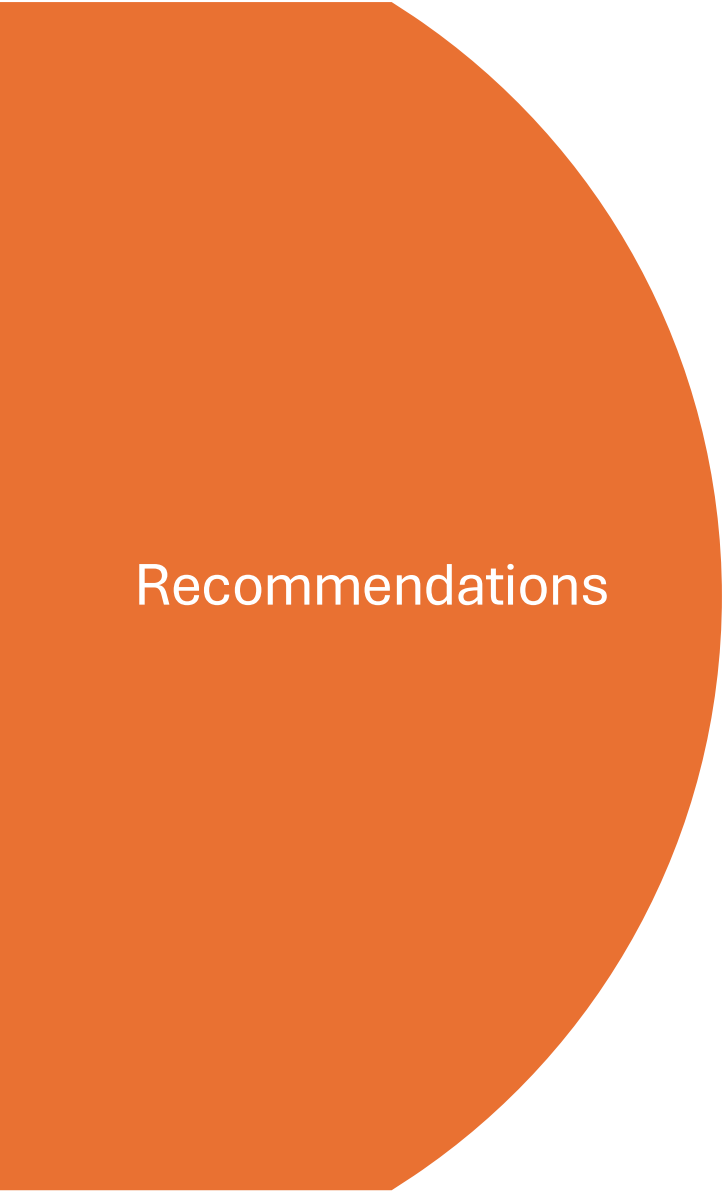
- Current findings are preliminary and require confirmation with a larger sample.

# Analysis Summary

- The analysis identified **Scratch** and **Leak** as the most common defect types, accounting for **66.7%** of all recorded defects. Among the potential contributing factors, **operator experience** showed the strongest association with defect occurrence, with new operators exhibiting a defect rate of **33.3%**, compared with **11.4%** for experienced operators. In comparison, differences in defect rates between **Jetline-1 and Jetline-2 fixtures (20.0% vs. 16.0%)** and between the **Day and Night shifts (17.65% vs. 18.75%)** were relatively small, suggesting these factors had less influence on defect occurrence during the data collection period. Based on these findings, improvement efforts should prioritize reducing **Scratch** and **Leak** defects while strengthening training, onboarding, and standardized work practices for new operators.

# Recommendations

| Recommendation                                                                           | Root cause targeted                               | Supporting analysis                                         | Control                                                  |
|------------------------------------------------------------------------------------------|---------------------------------------------------|-------------------------------------------------------------|----------------------------------------------------------|
| Structured OJT, supervised builds, competency test, and approval before independent work | <b>X1: Operator experience</b>                    | New operators: 33.3%; experienced: 11.4%; risk ratio = 2.92 | Certification and weekly new-operator defect-rate review |
| Include scratch-safe handling in the OJT standard                                        | <b>X1: Operator experience</b> and Pareto finding | Scratch was 44.4% of recorded defects                       | Weekly scratch Pareto and handling audit                 |
| Include seal inspection, fitting verification, and torque steps in the OJT checklist     | <b>X1: Operator experience</b> and Pareto finding | Leak was 22.2% of recorded defects                          | Daily leak-test audit                                    |
| Continue routine fixture inspection without prioritizing fixture replacement             | <b>X2: Fixture</b>                                | X2 was not statistically significant, (p=1.000)             | Monthly fixture inspection                               |
| Maintain consistent procedures across both shifts; continue collecting data              | <b>X3: Shift</b>                                  | X3 was not statistically significant, (p=1.000)             | Weekly shift-level defect-rate review                    |

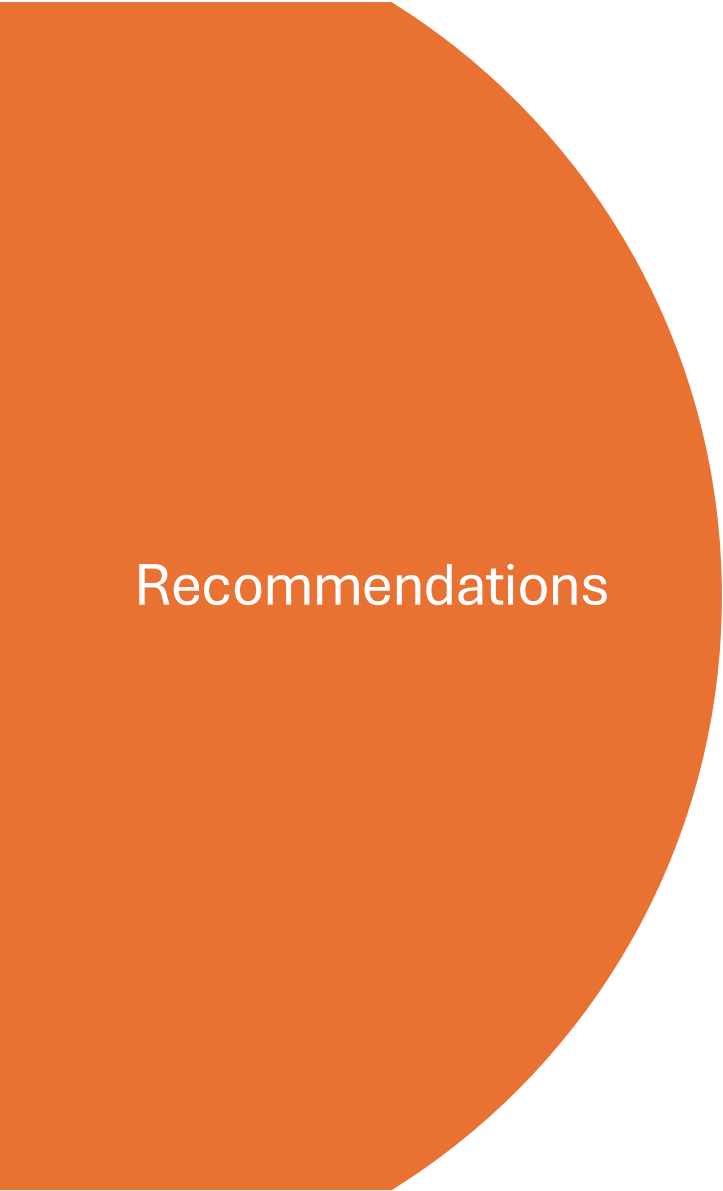
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## Recommendations

### Reduce Scratch Defects

- **Recommendation**
  - Improve material handling procedures and protective handling methods.
  - Reinforce work instructions to prevent scratches during assembly.
- **Rationale**
  - Scratch accounted for **44.4%** of all defects.
- **Control**
  - Monitor scratch defects using weekly Pareto charts.

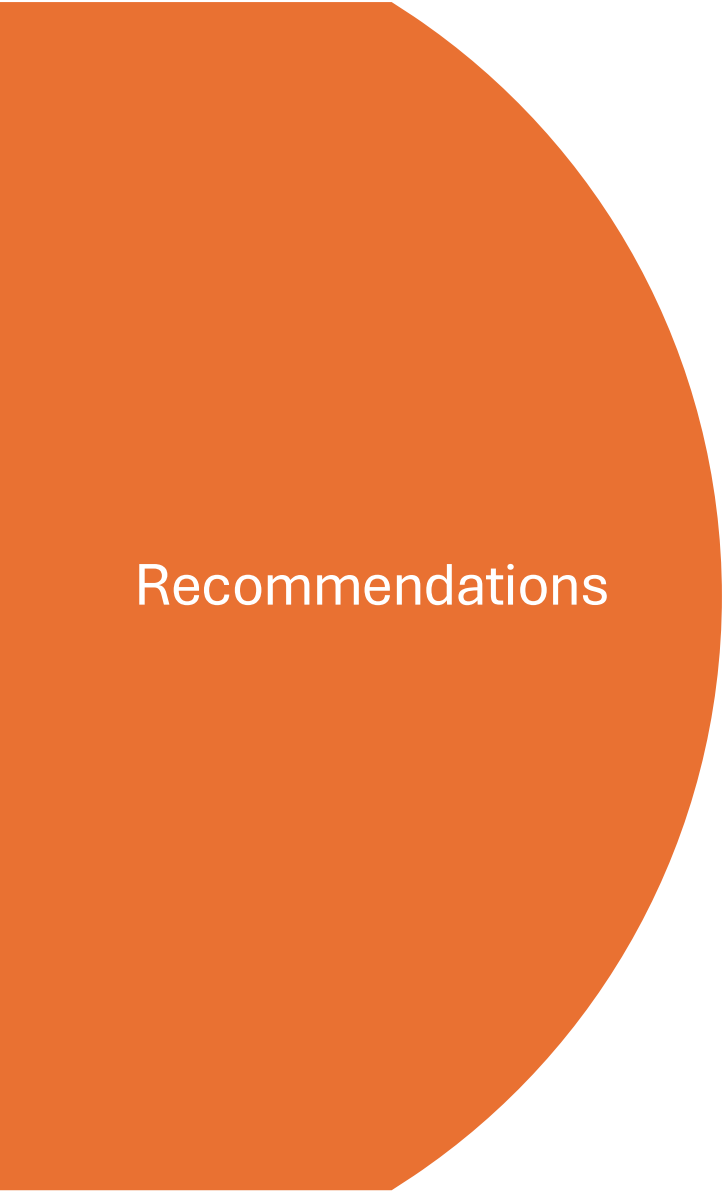




## Recommendations

### Reduce Leak Defects

- **Recommendation**
    - Standardize inspection of seals and fittings.
    - Verify assembly torque and leak testing procedures.
  - **Rationale**
    - Leak represented **22.2%** of recorded defects.
  - **Control**
    - Conduct daily leak-test verification.
    - Track leak defects monthly.
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## Recommendations

### Improve Preventive Maintenance

- **Recommendation**

- Schedule routine inspection and alignment verification of Jetline fixtures.

- **Rationale**

- Jetline-1 exhibited a slightly higher defect rate (20% vs.16%).

- **Control**

- Maintain preventive maintenance logs.
  - Audit fixture alignment monthly.
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# Control Plan Slide

| Improvement           | Control                           | Frequency | Owner                 |
|-----------------------|-----------------------------------|-----------|-----------------------|
| New operator training | Monitor defect rate by experience | Monthly   | Production Supervisor |
| Scratch reduction     | Weekly Pareto review              | Weekly    | Quality Engineer      |
| Leak reduction        | Leak test audit                   | Daily     | Quality Inspector     |
| Fixture maintenance   | Preventive maintenance schedule   | Monthly   | Maintenance Team      |

# Target and Reaction Plan

| Metric                   | Target         | Frequency | Owner                 | Reaction plan                                                  |
|--------------------------|----------------|-----------|-----------------------|----------------------------------------------------------------|
| Overall defect rate      | <8%            | Weekly    | Quality engineer      | If above 8%, contain affected tanks and review within 24 hours |
| New-operator defect rate | <10%           | Weekly    | Production supervisor | Return operator to supervised work                             |
| Scratch rate             | ≤2%            | Weekly    | Quality engineer      | Inspect handling points and retrain                            |
| Leak rate                | ≤1%            | Daily     | Quality inspector     | Stop release, verify torque/seals, and retest                  |
| Fixture inspection       | 100% completed | Monthly   | Maintenance lead      | Complete inspection before further production                  |

# Project Conclusion

- The project established an 18% baseline defect rate and identified Scratch and Leak as the leading defect categories. Operator experience showed the largest observed association, although the result was not statistically significant. The recommended actions will be piloted and evaluated against the less-than-8% target. Effectiveness will be confirmed through continued data collection and post-improvement analysis.

